

Physical Chemistry (II)

Lecture 05

Particle in a Box



In the Last Two Classes

1. Wavefunction and its interpretation

2. Quantum measurement

- Observables and operators
- Measurement and collapse of wave function
- Uncertainty principle

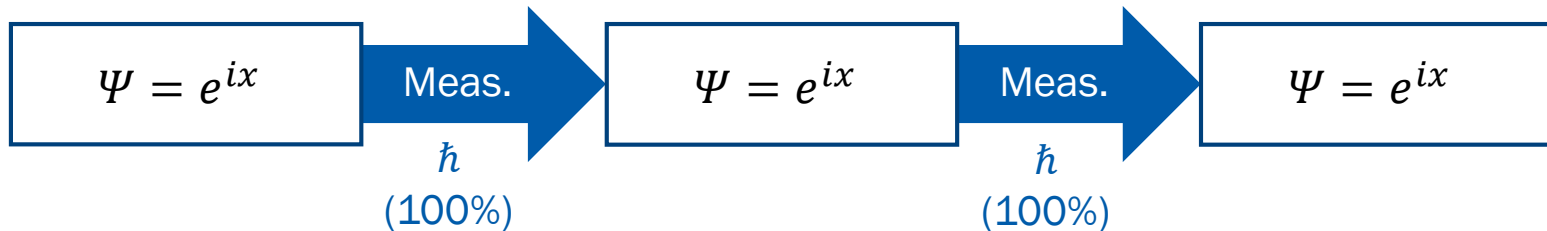
3. Time evolution of wavefunction: Schrödinger equation

❖ Example: 1D particle in a box

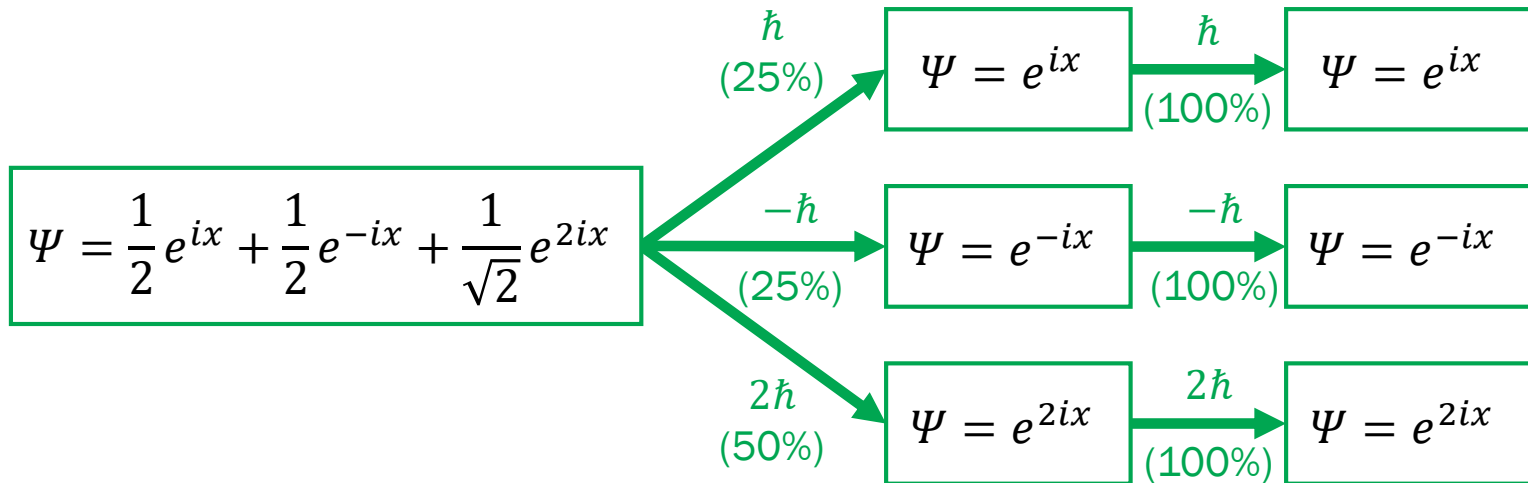


Quantum Measurement

- If the wavefunction is the eigenfunction of the operator



- Otherwise



Expectation Values

For a wavefunction Ψ , the expectation value of A is

$$\langle A \rangle = \int \Psi^* \hat{A} \Psi \, d\tau$$

Also,

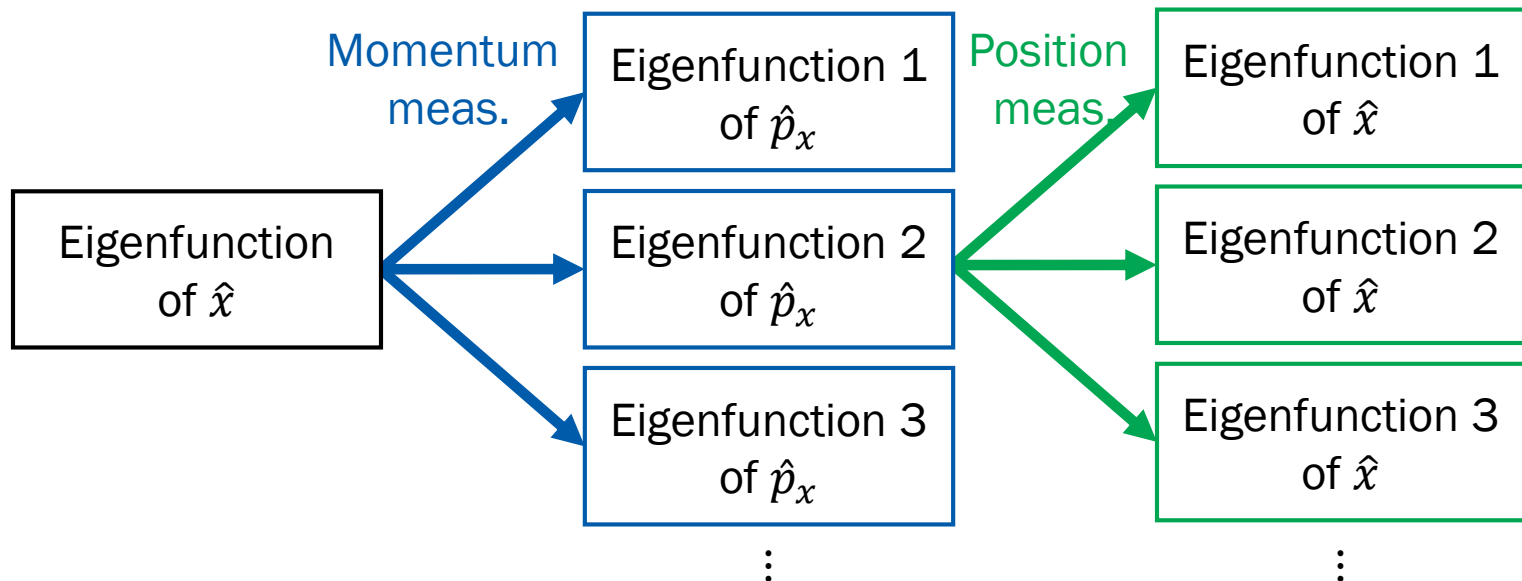
$$\langle A^2 \rangle = \int \Psi^* \hat{A} \hat{A} \Psi \, d\tau$$

- Variance $\sigma_A^2 = \langle A^2 \rangle - \langle A \rangle^2$
- Standard deviation $\sigma_A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$

Case of Two Observables

The eigenfunction of one QM operator may not be the eigenfunction of another QM operator

(e.g. eigenfunction of $\hat{x} \neq$ eigenfunction of $\hat{p}_x$)



Last class

Generalized Uncertainty Principle

$$\sigma_A^2 \sigma_B^2 \geq -\frac{1}{4} \langle [\hat{A}, \hat{B}] \rangle^2$$

e.g. position-momentum uncertainty

$$[\hat{x}, \hat{p}_x] = \hat{x}\hat{p}_x - \hat{p}_x\hat{x} = x \frac{\hbar}{i} \frac{\partial}{\partial x} - \frac{\hbar}{i} \frac{\partial}{\partial x} x = i\hbar$$

$$\sigma_x^2 \sigma_{p_x}^2 \geq -\frac{1}{4} \langle i\hbar \rangle^2 = \left(\frac{\hbar}{2} \right)^2$$

$$\sigma_x \sigma_{p_x} \geq \frac{\hbar}{2}$$

Last class

Time Evolution of Wavefunction

$$i\hbar \frac{\partial}{\partial t} \Psi(x, y, z, t) = \hat{H} \Psi(x, y, z, t)$$

where

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(\hat{x}, \hat{y}, \hat{z}, t)$$

(Time-dependent Schrödinger equation)

Time-Independent Potential

- Time-independent Schrödinger equation

$$\hat{H}\psi(x, y, z) = E\psi(x, y, z)$$

→ Eigenvalue equation for the Hamiltonian

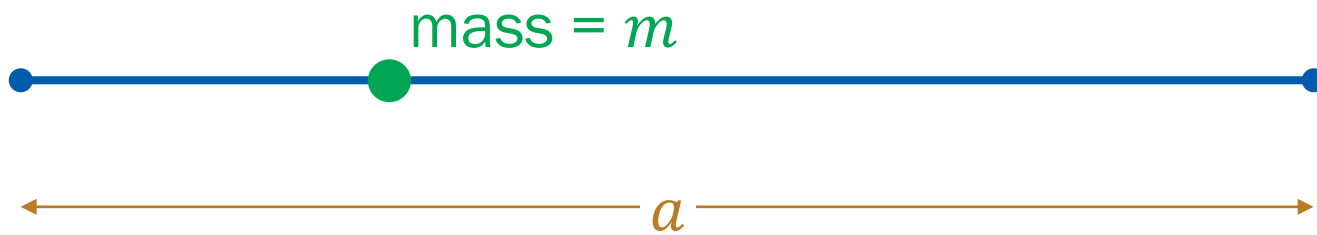
- Temporal solution: $T(t) = e^{-iEt/\hbar}$

$$|T(t)|^2 = 1$$

$$|\Psi(x, y, z, t)|^2 = |\psi(x, y, z)|^2$$

Last class

Particle in a 1D Box



$$V(x) = \begin{cases} \infty, & x < 0 \\ 0, & 0 \leq x \leq a \\ \infty, & x > a \end{cases}$$

Final Solutions

- Index $n = 1, 2, 3, \dots$ (“quantum number”)
- Time-independent wavefunction

$$\psi_n(x) = \sqrt{2/a} \sin\left(\frac{n\pi x}{a}\right), \quad 0 \leq x \leq a$$

- Energy

$$E_n = \frac{h^2}{8ma^2} n^2$$

- Total wavefunction

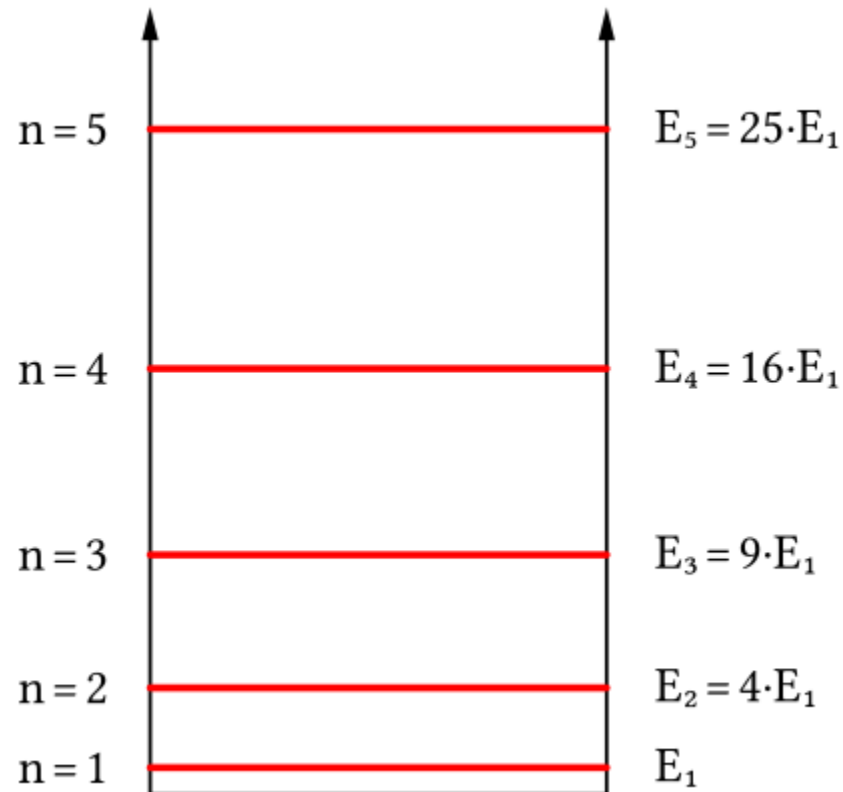
$$\Psi_n(x, t) = \psi_n(x)T_n(t) = \sqrt{2/a} \sin\left(\frac{n\pi x}{a}\right) e^{-iE_n t/\hbar}$$

RS 3.1

M 3.5

Energy Levels

$$E_n = \frac{h^2}{8ma^2} n^2$$



Wavefunctions: $n = 1$ State

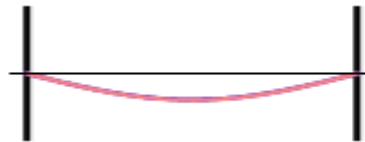
If the system is at state $n = 1$, the wavefunctions are

$$\psi_1(x) = \sqrt{2/a} \sin\left(\frac{\pi x}{a}\right)$$

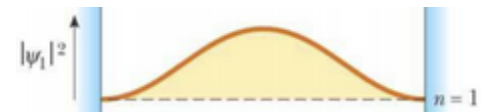
$$\Psi_1(x, t) = \psi_1(x) e^{-iE_1 t/\hbar} = \sqrt{2/a} \sin\left(\frac{\pi x}{a}\right) e^{-i\pi \hbar t/4ma^2}$$



$\psi_1(x)$



$\Psi_1(x, t)$



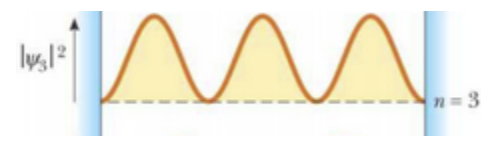
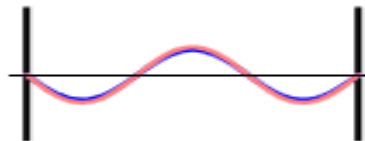
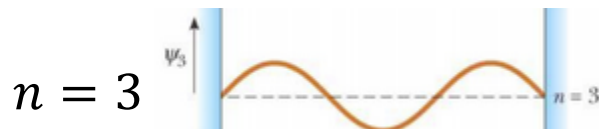
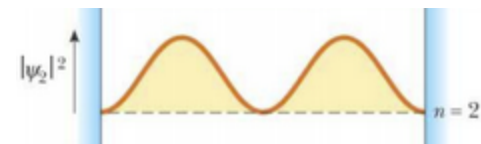
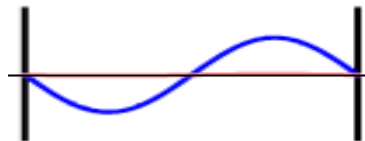
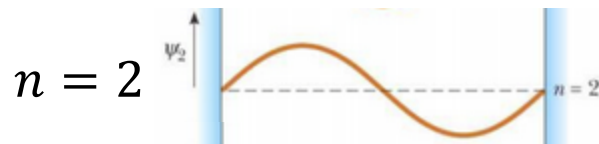
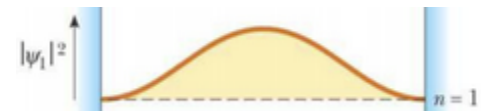
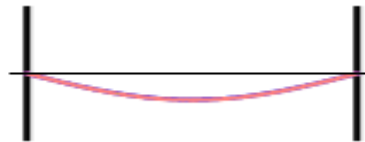
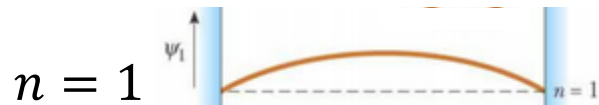
$$|\Psi_1(x, t)|^2 = |\psi_1(x)|^2$$

Wavefunctions

$$\psi_n(x)$$

$$\Psi_n(x, t)$$

$$|\Psi_n(x, t)|^2 = |\psi_n(x)|^2$$



Let's Practice QM

1. Wavefunction and its interpretation: probability
2. Quantum measurement: expectation values, uncertainty
3. Time evolution of wavefunction

Take the $n = 1$ state as an example

Probability

The probability that the particle is found between $x = 0$ and $a/2$ is

$$\begin{aligned} P(0, a/2) &= \int_0^{a/2} |\Psi_1(x, t)|^2 dx = \int_0^{a/2} |\psi_1(x)|^2 dx \\ &= \int_0^{a/2} \left(\frac{2}{a}\right) \sin^2\left(\frac{\pi x}{a}\right) dx = \frac{1}{2} \end{aligned}$$

Expectation Values

The expectation value of observable A is

$$\begin{aligned}\langle A \rangle &= \int_{-\infty}^{\infty} \psi_1^*(x) \hat{A} \psi_1(x) dx \\ &= \int_0^a \sqrt{2/a} \sin\left(\frac{\pi x}{a}\right) \hat{A} \left[\sqrt{2/a} \sin\left(\frac{\pi x}{a}\right) \right] dx\end{aligned}$$

Expectation Value of Position

$$\langle x \rangle = \frac{2}{a} \int_0^a \sin\left(\frac{\pi x}{a}\right) x \sin\left(\frac{\pi x}{a}\right) dx = \frac{2}{a} \int_0^a x \sin^2\left(\frac{\pi x}{a}\right) dx$$

$$\therefore \langle x \rangle = \frac{a}{2}$$



Expectation Value of Momentum

$$\begin{aligned}
 \langle p_x \rangle &= \frac{2}{a} \int_0^a \sin\left(\frac{\pi x}{a}\right) \left(\frac{\hbar}{i} \frac{\partial}{\partial x} \right) \sin\left(\frac{\pi x}{a}\right) dx \\
 &= \frac{2\hbar}{ia} \int_0^a \sin\left(\frac{\pi x}{a}\right) \left(\frac{\pi}{a} \right) \cos\left(\frac{\pi x}{a}\right) dx \\
 &= \frac{2\pi\hbar}{ia^2} \int_0^a \sin\left(\frac{\pi x}{a}\right) \cos\left(\frac{\pi x}{a}\right) dx \\
 &\quad \therefore \langle p_x \rangle = 0
 \end{aligned}$$

Variances

Similarly,

$$\langle x^2 \rangle = \frac{a^2}{3} - \frac{a^2}{2\pi^2}; \quad \langle p_x^2 \rangle = \frac{h^2}{4a^2}$$

which leads to

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{a^2}{3} - \frac{a^2}{2\pi^2} - \left(\frac{a}{2}\right)^2 = \frac{a^2}{12} - \frac{a^2}{2\pi^2}$$

$$\sigma_{p_x}^2 = \langle p_x^2 \rangle - \langle p_x \rangle^2 = \frac{h^2}{4a^2}$$

Uncertainty Principle

Now,

$$\sigma_x^2 \sigma_{p_x}^2 = \left(\frac{a^2}{12} - \frac{a^2}{2\pi^2} \right) \frac{h^2}{4a^2} = h^2 \left(\frac{1}{48} - \frac{1}{8\pi^2} \right) = 0.00817h^2$$

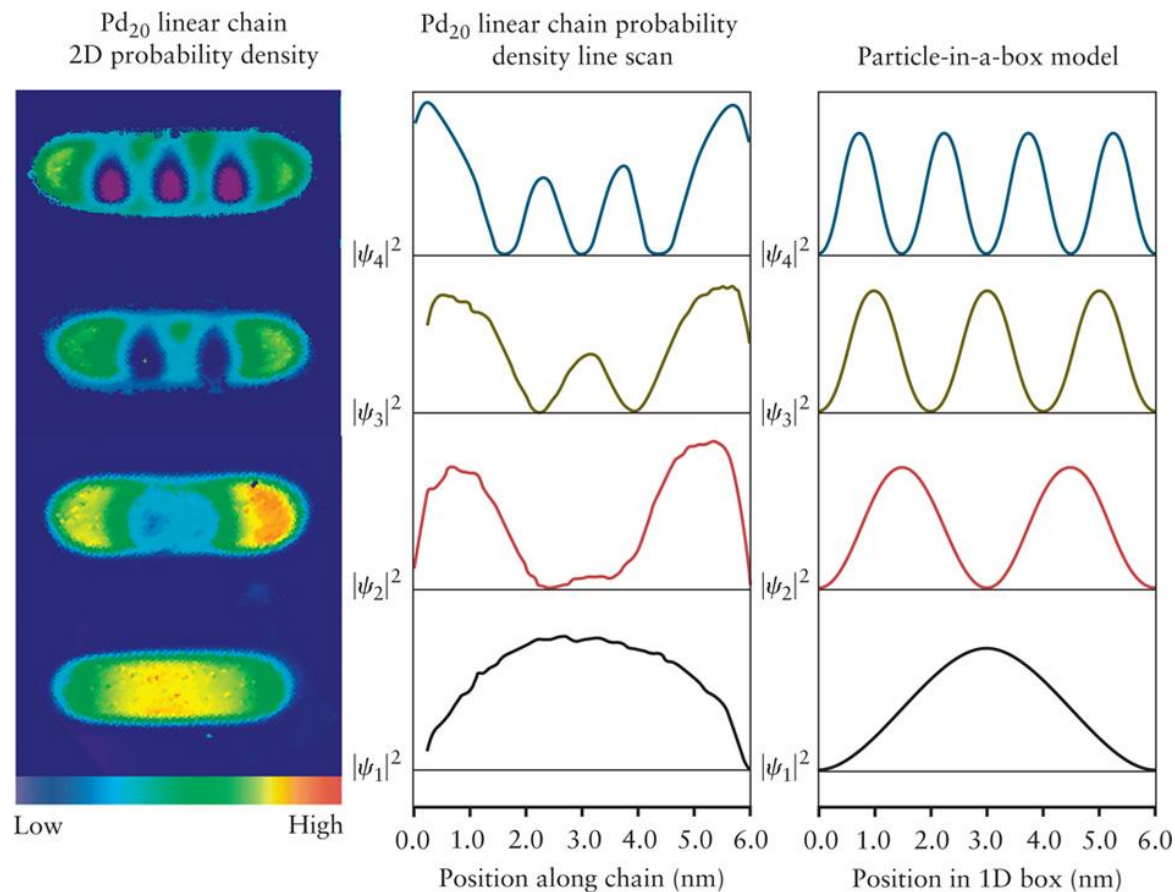
$$\sigma_x \sigma_{p_x} = 0.090h$$

The uncertainty principle states that

$$\sigma_x \sigma_{p_x} \geq h/4\pi = 0.080h$$

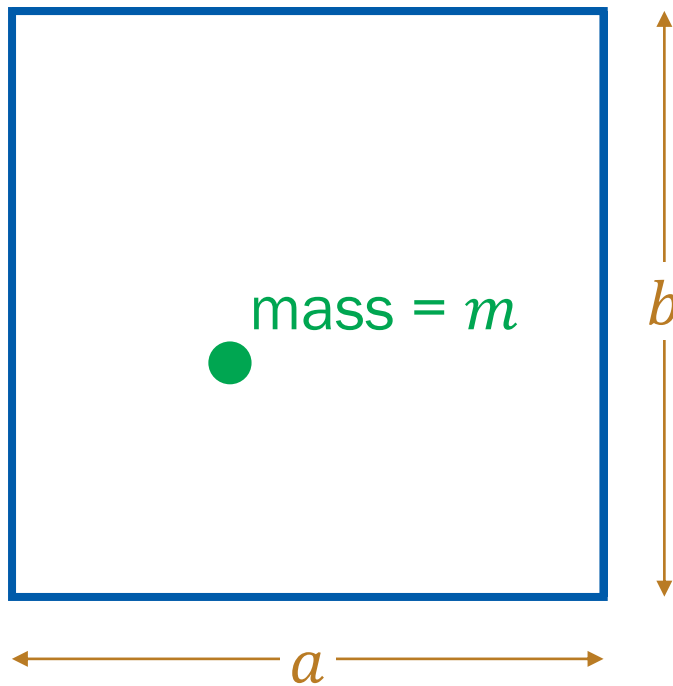
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Application: Pd Chain



(Image: Oxtoby et al., *Principles of Modern Chemistry*, 7th Ed. (2011), Fig. 4.27)

2D Particle in a Box



$$\mathcal{V}(x, y) = \mathcal{V}_x(x)\mathcal{V}_y(y)$$

$$\mathcal{V}_x(x) = \begin{cases} \infty, & x < 0 \\ 0, & 0 \leq x \leq a \\ \infty, & x > a \end{cases}$$

$$\mathcal{V}_y(y) = \begin{cases} \infty, & y < 0 \\ 0, & 0 \leq y \leq b \\ \infty, & y > b \end{cases}$$

Time-Indep. Schrödinger Equation

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(x, y) \right] \psi(x, y) = E\psi(x, y)$$

For $0 \leq x \leq a$ and $0 \leq y \leq b$,

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \psi(x, y) = E\psi(x, y)$$

Let $\psi(x, y) = X(x)Y(y)$; rearrangement leads to

$$-\frac{\hbar^2}{2m} \left(\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} \right) = E$$

Separation of Variables

$$-\frac{\hbar^2}{2m} \left(\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} \right) = E$$

$$\underbrace{-\frac{\hbar^2}{2m} \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2}}_{\text{function of } y} = E + \underbrace{\frac{\hbar^2}{2m} \frac{1}{X} \frac{\partial^2 X}{\partial x^2}}_{\text{function of } x} = E_y$$

System of Equations

$$-\frac{\hbar^2}{2m} \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = E - E_y = E_x$$

$$-\frac{\hbar^2}{2m} \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = E_y$$

Each of these equations is identical with
the equation for a 1D particle in a box

Solutions

- Quantum numbers $n_x, n_y = 1, 2, 3, \dots$
- Energy levels

$$E_x = \frac{h^2}{8ma^2} n_x^2, \quad E_y = \frac{h^2}{8mb^2} n_y^2$$

$$E = E_x + E_y = \frac{h^2}{8m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} \right) = E_{(n_x, n_y)}$$

Solutions

- (Time-independent) Wavefunctions

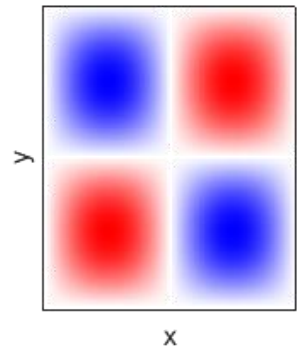
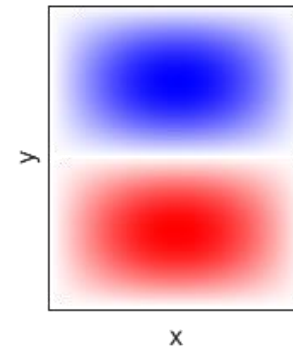
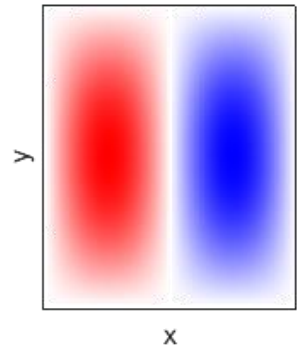
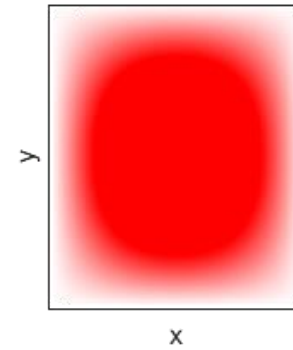
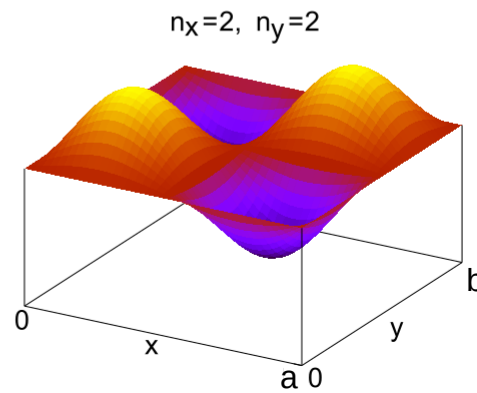
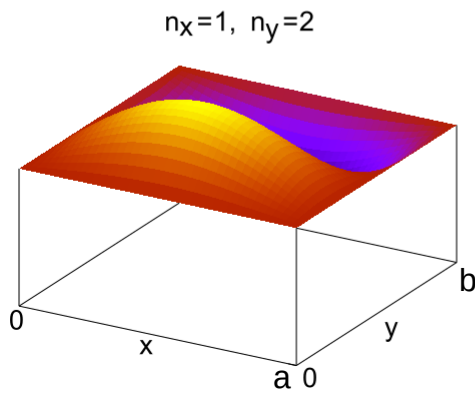
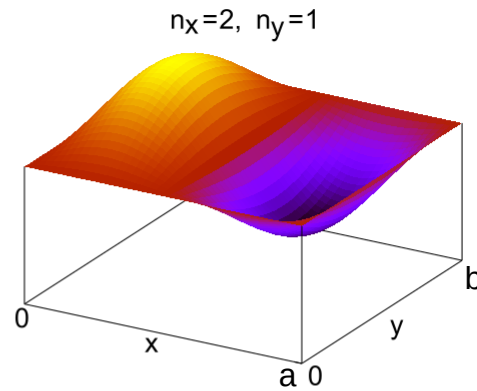
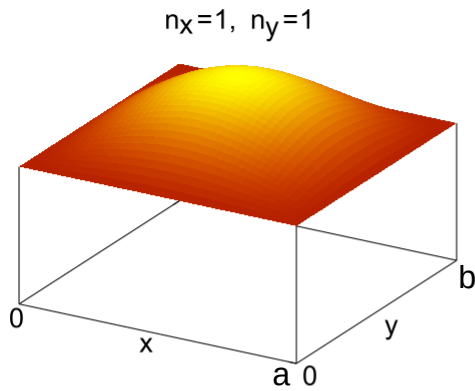
$$X(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n_x \pi x}{a}\right), \quad 0 \leq x \leq a$$

$$Y(y) = \sqrt{\frac{2}{b}} \sin\left(\frac{n_y \pi y}{b}\right), \quad 0 \leq y \leq b$$

$$\psi_{n_x, n_y}(x, y) = \sqrt{\frac{4}{ab}} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{b}\right)$$

A (area)

Wavefunctions

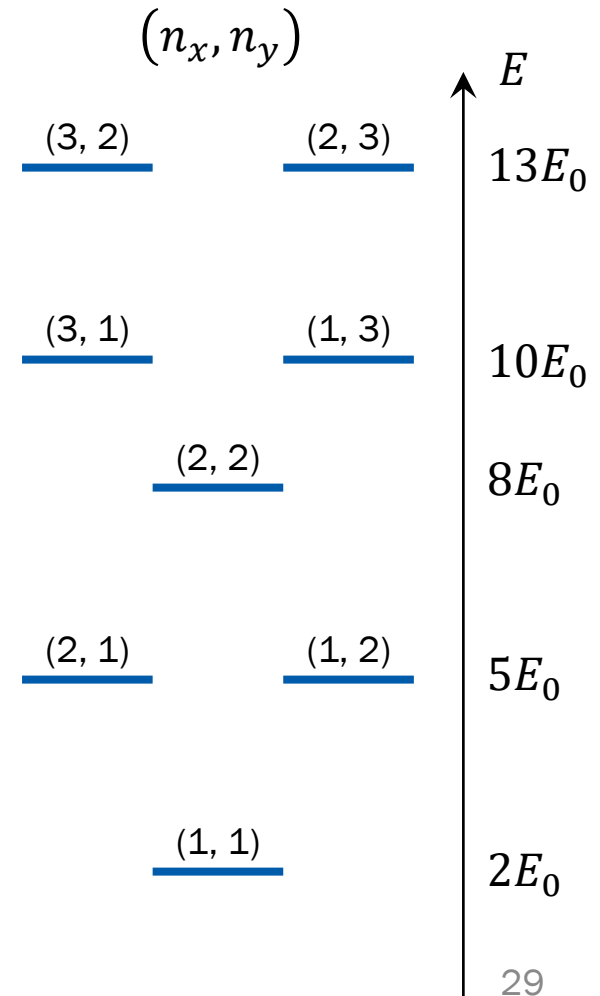


Degeneracy

When $a = b$,

$$E = \frac{h^2}{8ma^2} (n_x^2 + n_y^2) = E_0 (n_x^2 + n_y^2)$$

- **Degeneracy:** different states have the same energy
- The states are called **degenerate**



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When You Have Degeneracy

The **linear combination** of wavefunctions of degenerate states **also satisfies the Schrödinger equation**

For ψ_1 and ψ_2 such that $\hat{H}\psi_1 = E_0\psi_1$ and $\hat{H}\psi_2 = E_0\psi_2$, we can define $\phi = c_1\psi_1 + c_2\psi_2$ and obtain

$$\hat{H}\phi = \hat{H}(c_1\psi_1 + c_2\psi_2) = c_1E_0\psi_1 + c_2E_0\psi_2 = E_0\phi$$

Example

The states (2, 1) and (1, 2) are degenerate (energy = $5E_0$)

$$\psi_{2,1}(x, y) = \frac{2}{a} \sin\left(\frac{2\pi x}{a}\right) \sin\left(\frac{\pi y}{a}\right)$$

$$\psi_{1,2}(x, y) = \frac{2}{a} \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{2\pi y}{a}\right)$$

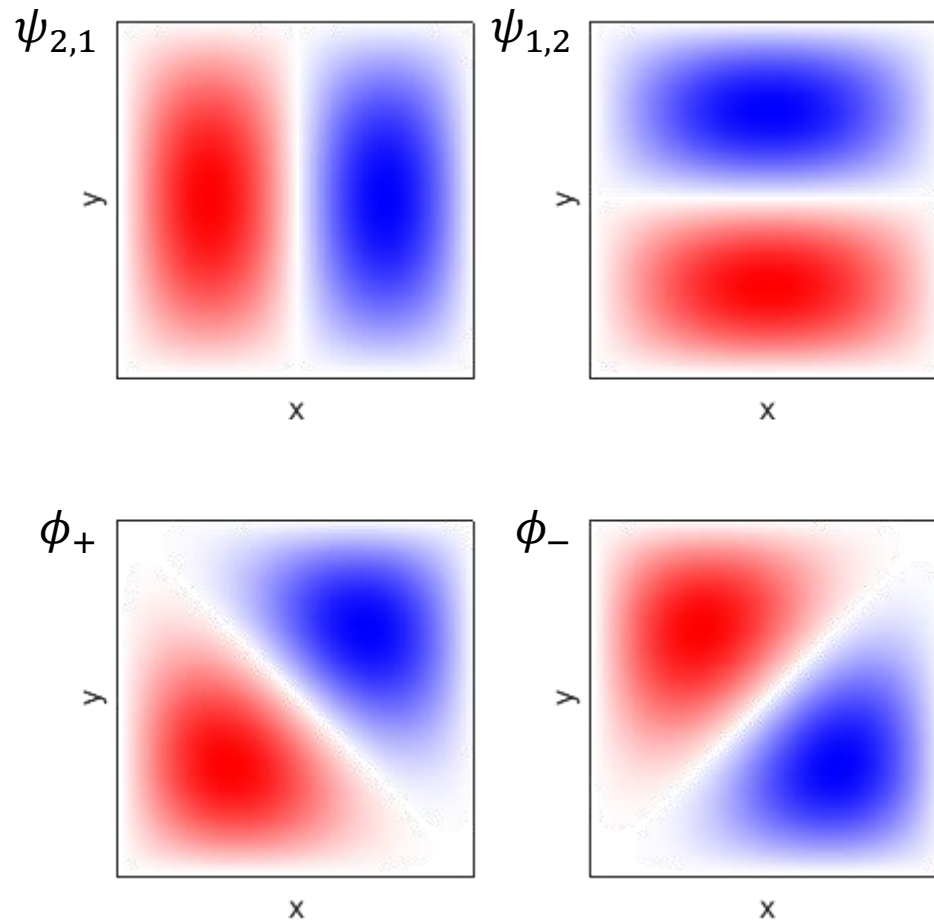
We can make

$$\phi_+ = \frac{1}{\sqrt{2}}(\psi_{2,1} + \psi_{1,2}); \quad \phi_- = \frac{1}{\sqrt{2}}(\psi_{2,1} - \psi_{1,2})$$

 which are also the solutions of the Schrödinger equation

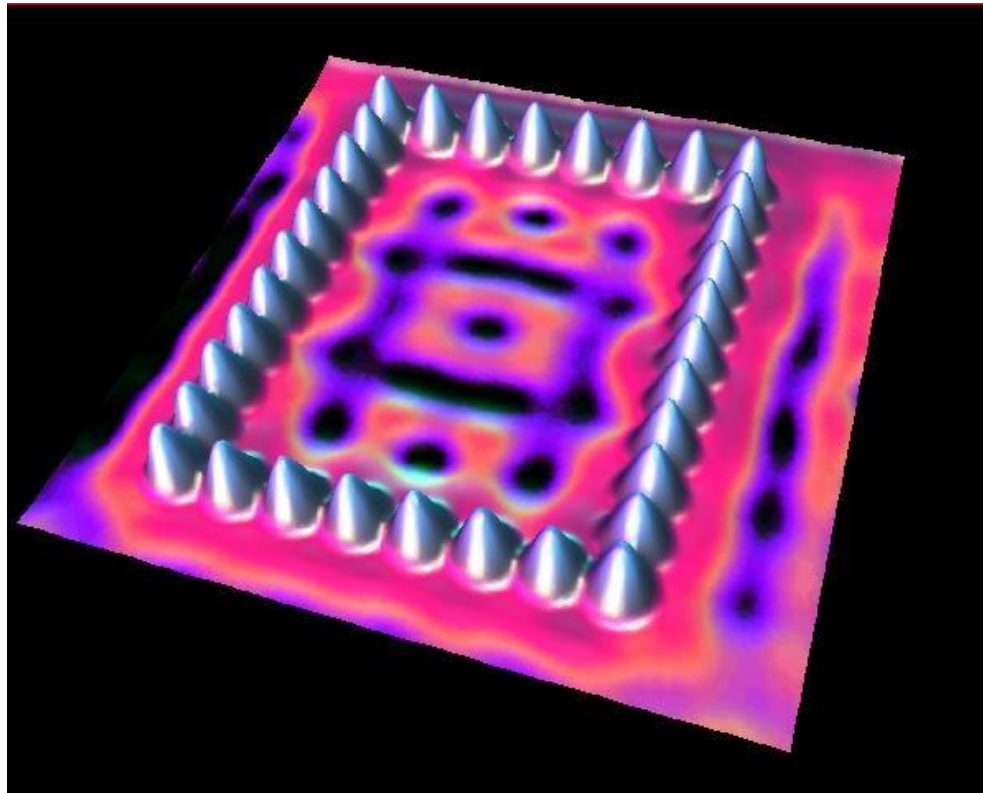
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Example



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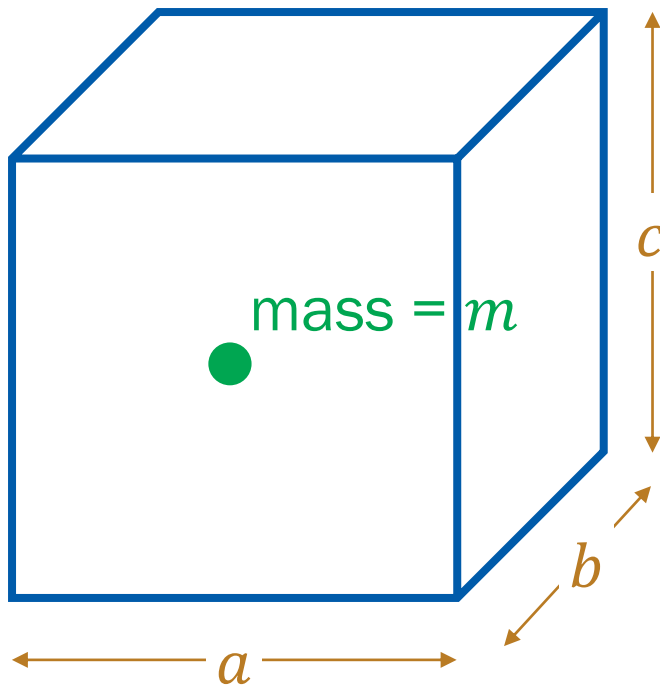
Application: “Quantum Square”



STM image for Fe atoms on a Cu(111) surface

(Image: <https://www.pinterest.co.kr/pin/469641067361529171/>)

3D Particle in a Box



$$\mathcal{V}(x, y, z) = \mathcal{V}_x(x)\mathcal{V}_y(y)\mathcal{V}_z(z)$$

$$\mathcal{V}_x(x) = \begin{cases} \infty, & x < 0 \\ 0, & 0 \leq x \leq a \\ \infty, & x > a \end{cases}$$
$$\mathcal{V}_y(y) = \begin{cases} \infty, & y < 0 \\ 0, & 0 \leq y \leq b \\ \infty, & y > b \end{cases}$$
$$\mathcal{V}_z(z) = \begin{cases} \infty, & z < 0 \\ 0, & 0 \leq z \leq c \\ \infty, & z > c \end{cases}$$

Time-Indep. Schrödinger Equation

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(x, y, z) \right] \psi(x, y, z) = E\psi(x, y, z)$$

For $0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$,

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi(x, y, z) = E\psi(x, y, z)$$

Let $\psi(x, y, z) = X(x)Y(y)Z(z)$; rearrangement leads to

$$-\frac{\hbar^2}{2m} \left(\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} + \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} \right) = E$$

Separation of Variables

$$-\frac{\hbar^2}{2m} \left(\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} + \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} \right) = E$$

$$\underbrace{-\frac{\hbar^2}{2m} \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2}}_{\text{function of } z} = E + \underbrace{\frac{\hbar^2}{2m} \left(\frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} \right)}_{\text{function of } x, y} = E_z$$

Similarly, the right-handed side gives

$$-\frac{\hbar^2}{2m} \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = (E - E_z) + \frac{\hbar^2}{2m} \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = E_y$$

System of Equations

$$-\frac{\hbar^2}{2m} \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = E - E_y - E_z = E_x$$

$$-\frac{\hbar^2}{2m} \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = E_y$$

$$-\frac{\hbar^2}{2m} \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} = E_z$$

Each of these equations is identical with the equation
for a 1-dimensional particle in a box

Solutions

- Quantum numbers $n_x, n_y, n_z = 1, 2, 3, \dots$
- Energy levels

$$E_x = \frac{h^2}{8ma^2} n_x^2, \quad E_y = \frac{h^2}{8mb^2} n_y^2, \quad E_z = \frac{h^2}{8mc^2} n_z^2$$

$$E = E_x + E_y + E_z = \frac{h^2}{8m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2} \right) = E_{(n_x, n_y, n_z)}$$

Solutions

- Wavefunctions

$$X(x) = \sqrt{2/a} \sin\left(\frac{n_x \pi x}{a}\right), \quad 0 \leq x \leq a$$

$$Y(y) = \sqrt{2/b} \sin\left(\frac{n_y \pi y}{b}\right), \quad 0 \leq y \leq b$$

$$Z(z) = \sqrt{2/c} \sin\left(\frac{n_z \pi z}{c}\right), \quad 0 \leq z \leq c$$

$$\psi(x, y, z) = \sqrt{\frac{8}{abc}} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{b}\right) \sin\left(\frac{n_z \pi z}{c}\right)$$

V (volume)

Degeneracy

$$E_{(n_x, n_y, n_z)} = \frac{h^2}{8m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2} \right)$$

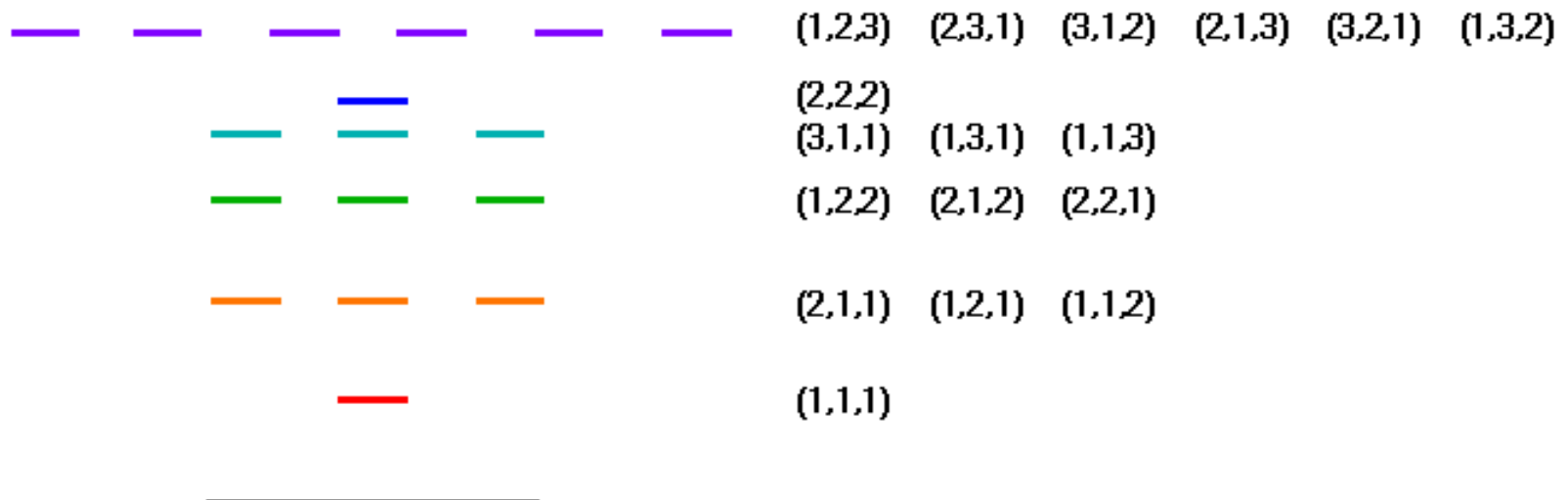
If $a = b = c$,

$$E_{(n_x, n_y, n_z)} = \frac{h^2}{8ma^2} (n_x^2 + n_y^2 + n_z^2)$$

Different states (= different combinations of n_x, n_y, n_z) can have the same energy → **degenerate states**

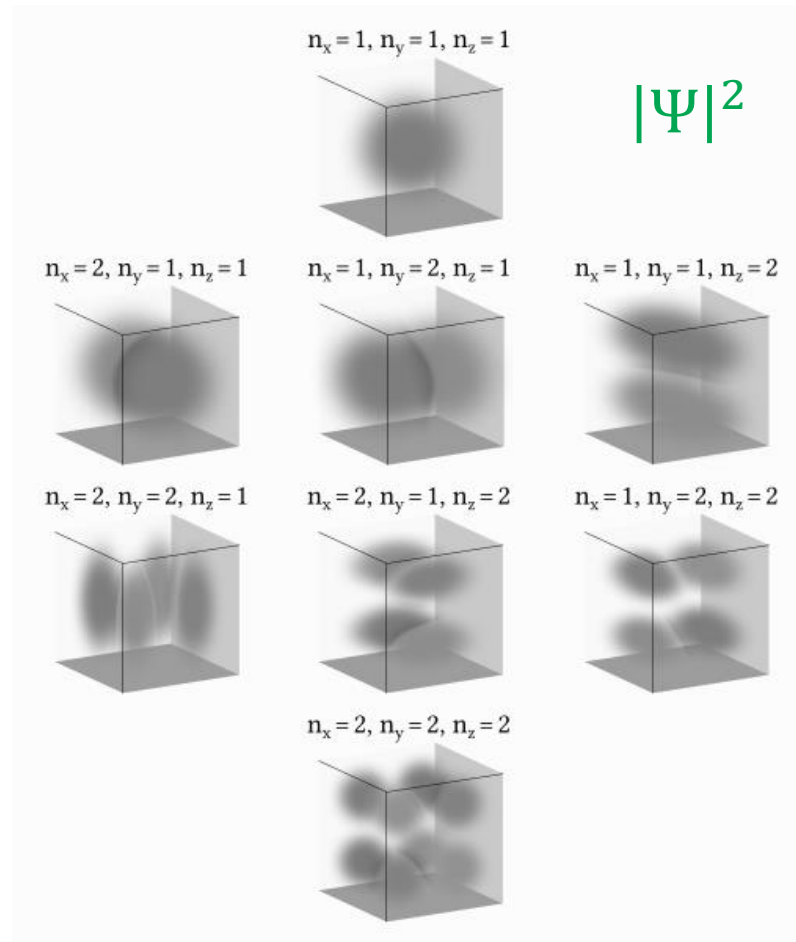
Energy Levels

$$E_{(n_x, n_y, n_z)} = \frac{h^2}{8ma^2} (n_x^2 + n_y^2 + n_z^2)$$



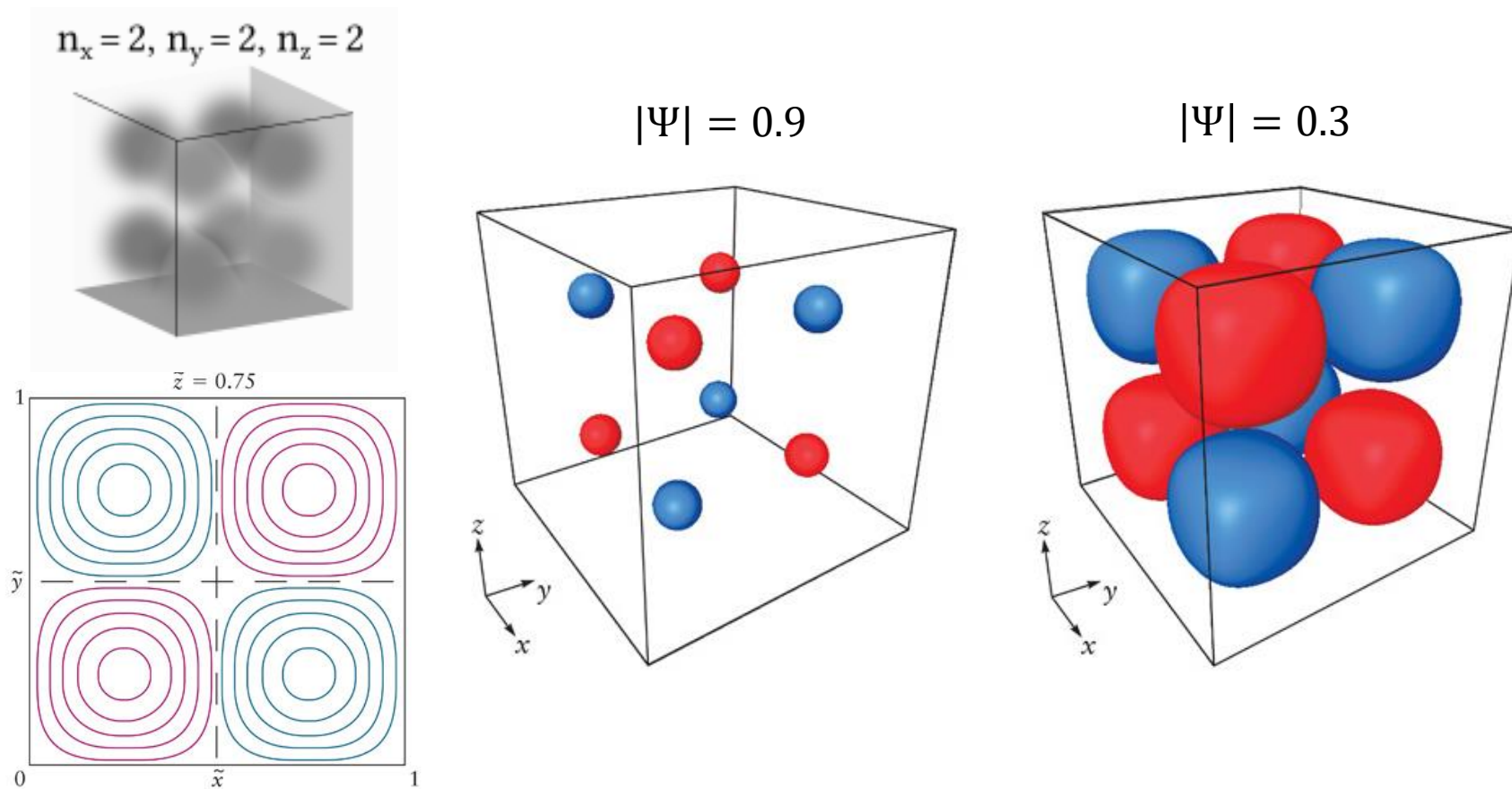
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How to Visualize 3D WF?



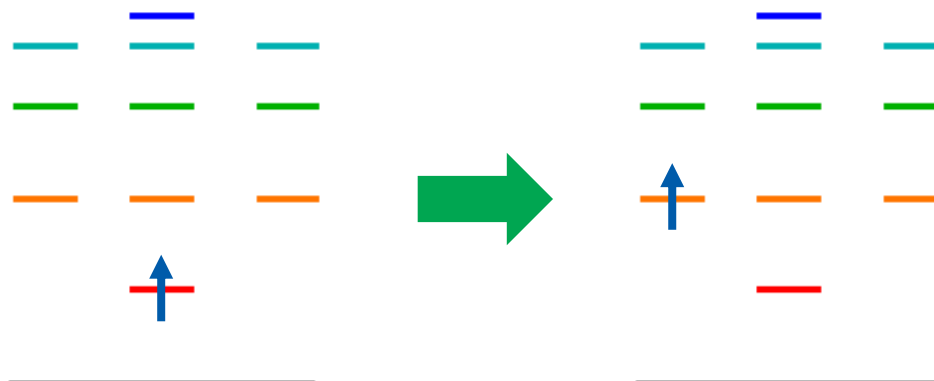
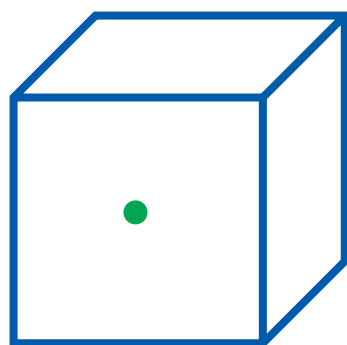
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Contour Surfaces



Not in text

Application: Molecular Translation



$$\Delta E = \frac{h^2}{8ma^2} (6 - 3) = \frac{3h^2}{8ma^2} = \frac{hc}{\lambda}$$

For a He atom (6.646×10^{-27} kg) in a box with $a = 0.1$ m,

$$\Delta E = 2.48 \times 10^{-39} \text{ J} = 1.49 \times 10^{-18} \text{ kJ/mol}$$

$$\lambda = 8.02 \times 10^{13} \text{ m: undetectable!}$$